Quantum 2048

Planning with shared fate

Exact merge forecasts powered by Qiskit



Waterloo Qiskit Fall Fest · 2026
Local simulator · Python notebook + playable game

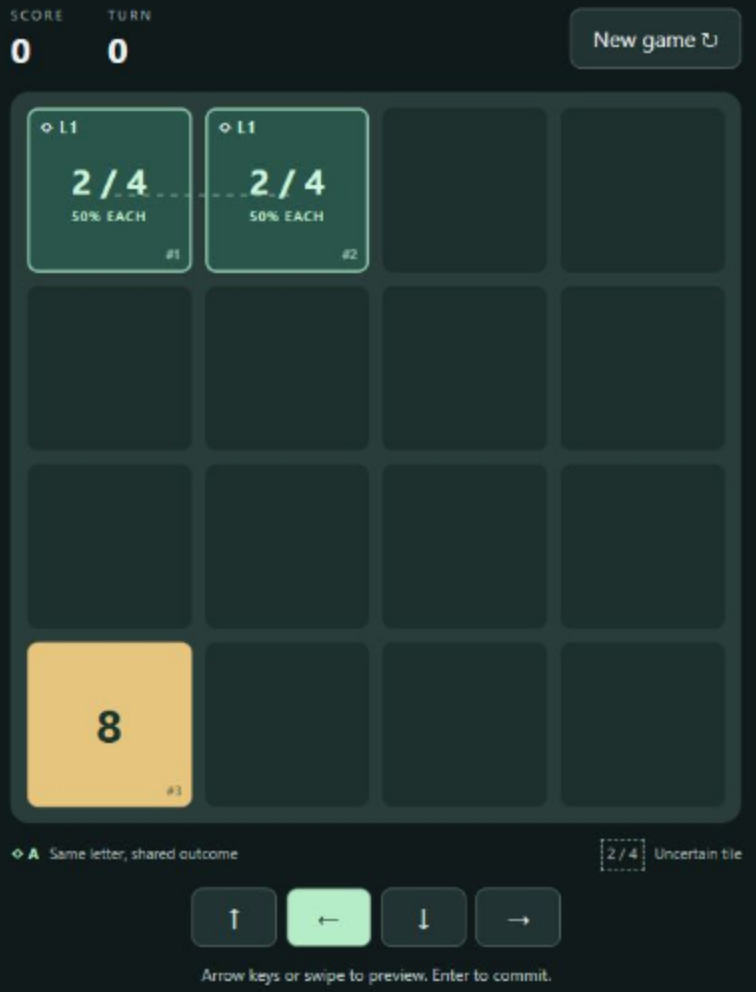
A forecast before every commitment

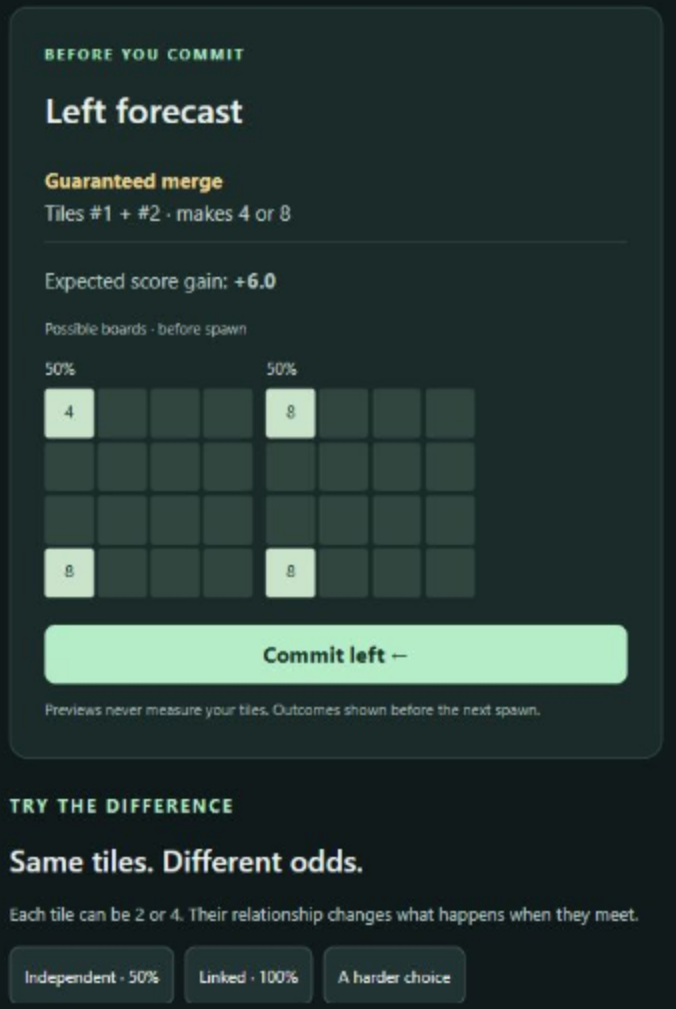
Choose a direction.
Read the chances.
Commit the move.

Lettered links work without color.

Outcome boards show movement and possible merges.

Measurement happens on contact.





TRY THE DIFFERENCE

Same tiles. Different odds.

Each tile can be 2 or 4. Their relationship changes what happens when they meet.

Independent · 50% | Linked · 100% | A harder choice

Same individual odds. Different joint outcomes.

INDEPENDENT

50%

Four equally likely outcomes
22 24 42 44

Expected score gain: +3

LINKED BELL PAIR

100%

Two equally likely outcomes
22 44

Expected score gain: +6

The cost of keeping a pair

Merge now

An outside contact can earn points.
It also reveals the distant partner.

Preserve the connection

Move the pair toward each other.
Spend board space and extra turns.

Tradeoff fixture: left gives expected +2 and resolves the pair.
Up gives +0 and preserves uncertainty before spawning.

Balance and player appeal still require real-user evaluation.

One Python core for the game and notebook

Qiskit

H and H–CX preparation
Exact basis probabilities
Persistent partial measurement

Shared game reducer

Clone and enumerate for preview
Measure on committed contact
Slide and merge by the same rules

Browser and notebook import the same model.
Four unresolved qubits keep forecasts small and local.

Executed evidence

198 / 400

Independent swipes merged
49.5% sampled · 50% exact

400 / 400

Linked swipes merged
100% sampled · 100% exact

Interference comparison

Coherent H–H: P(0) = 1
Dephased before the second H: P(0) = 0.5

Local ideal simulator results. 10 core tests pass.

Contribution and honest limits

The contribution

Understandable joint outcomes

Non-measuring visual forecasts

One reproducible technical artifact

The limits

Ideal simulator, no hardware run

Classically reproducible match odds

Usability and balance unvalidated

Notebook first. Playable demo alongside it.

Appendix · reliable demo and judge questions

90-second demo

Linked: guaranteed merge, commit
Independent: 50% forecast
Harder choice: compare left and up
Notebook: coherent vs dephased

Accurate short answers

Advantage? None demonstrated.
Entanglement proof? Z odds are insufficient.
Preview measurement? No.
Usability gains? Not yet measured.

Fallback: notebook sections 2–6 run without the server or browser.

Appendix · sources and submission readiness

Sources

uwquantum.com/qiskit-fall-fest
uwquantum.com/fall-fest-signup
quantum.cloud.ibm.com/docs/en/api/qiskit/quantum_info
github.com/nirbhaykwatra/Quantum2048
github.com/gabrielecirulli/2048

Human actions

Confirm cutoff, upload, talk length, AI/reuse rules and session time.
Add team identity, register individually, rehearse and present.
AI assisted this work. No source or assets copied from prior games.